

INS algorithm using quaternion model for low cost IMU

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Abstract

This paper presents a generic inertial navigation system (INS) error propagation model that does not rely on small misalignment angles assumption. The modelling uses quaternions in the computer frame approach. Based on this model, an INS algorithm is developed for low cost inertial measurement unit (IMU) to solve the initial attitudes uncertainty using in-motion alignment. The distribution approximation filter (DAF) is used to implement the non-linear data fusion algorithm.

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1. Introduction

Navigation systems are widely used to provide positioning information for field robotics, which is used to control the operating of the autonomous vehicles. Recent research on reliable and high integrity navigation systems for field robotics has become active in application areas such as mining, agriculture, construction and stevedoring [1,2].

Information inputs for the navigation systems are from navigation sensors. Two common broad categories of navigation sensors are dead reckoning sensors and external sensors [1,2]. Dead reckoning sensors provide robust and high frequency navigation data but accumulate errors with time, external sensors provide absolute information and bound navigation errors but output at low frequency [1–9]. The two types of sensors are usually integrated into one system to overcome their respective weaknesses and to fully utilise their strengths [1–8].

Among dead reckoning sensors, inertial measurement units (IMU) are widely used to construct inertial navigation system (INS). External navigation sensors are commonly used to aid INS [6–9]. An IMU usually contains a set of three orthogonal-installed accelerometers and three orthogonal-installed gyros. Fig. 1 illustrates an IMU installed in a vehicle. When the accelerometers and the gyros are directly installed in the vehicle body, the INS is called a strapdown INS. An inertial navigation system is a real time algorithm to calculate the position, velocity and attitude of the vehicle which carries the INS by integrating the acceleration and rotation rate signals from an IMU. The velocity and the position are calculated by double integration of the sum of the gravitational acceleration and the non-gravitational acceleration from the three accelerometers. By integrating the rotation rate

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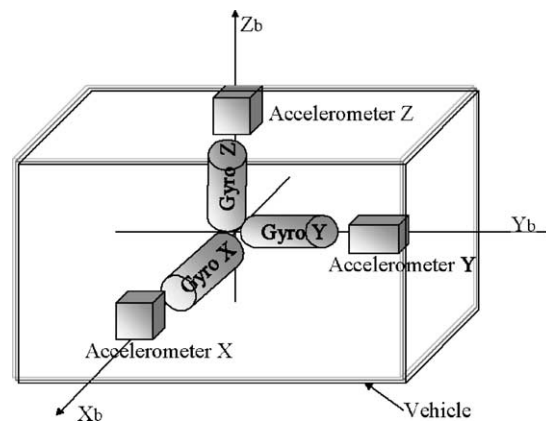


Fig. 1. An IMU is installed in a vehicle.

signals from three gyros, the angular orientation of the three accelerometers is determined. The calculated acceleration, velocity and position are transformed into the desired navigation coordinate system by this orientation information.

Fig. 2 shows the INS process. There are two main phases for INS operation: the alignment phase and the navigation phase. The navigation phase starts from the initial velocity, position and attitude. The process for determining these INS initial conditions is called alignment. Any errors in either the alignment phase or the navigation phase will be integrated and will propagate over time. These errors determine the performance and the navigation accuracy of the INS. INS error models serve for the error analysis and the implementation of a data fusion filter in the INS algorithms [10]. In the literature there are two approaches to the derivation of INS error models [11]: the phi-angle approach (or the true frame approach) and the psi-angle approach (or the computer approach). It has been shown that the models from these two approaches are equivalent and yield identical results [12–14]. Most

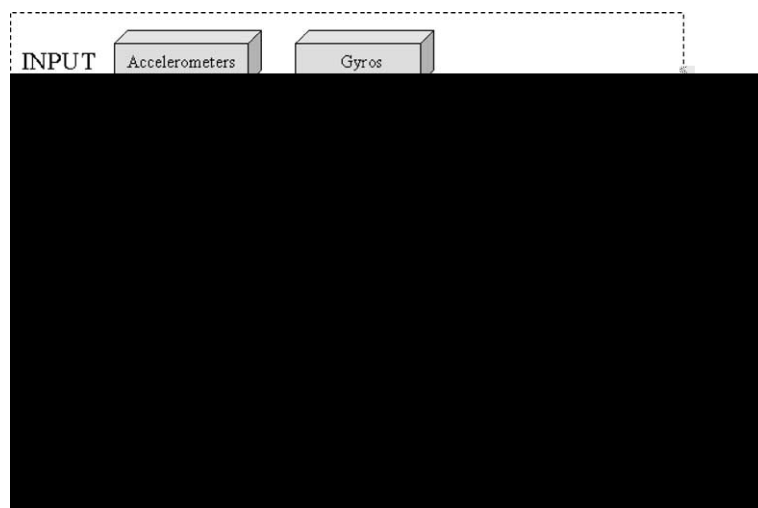


Fig. 2. INS process.

of the literature for INS modelling makes the assumption that all perturbation attitude errors are small angles. However, in many cases, this small angle assumption will not hold. For low cost inertial measurement whose sensitivity is not enough to measure the earth rate, the corresponding attitude error in both the phi-angle model and the psi-angle model could be up to $\pm 180^\circ$ making the small angle approximation not valid. Consequently, the INS error models with small angle assumption make the navigation system of low cost IMU inaccurate and unstable. This motivated the development of INS error model and navigation algorithm for low cost IMU in this paper.

More recently, attempts were made to model INS propagation errors with an unknown heading were presented. Pham introduced a Kalman filter mechanisation for the INS airstart system in 1992 [15]. This approach used two non-linear states to describe one heading angle. The attitude errors were not modelled. Stepanov et al. presented a model considering a large heading error and small tilt errors using the true frame [16]. Scherzinger and Reid developed a modified psi-angle model with arbitrary heading and small tilt errors using the psi-angle approach [17,18]. Four states are used to describe three psi-angles. The psi-angle Ψ_z which represents the correspondent heading error is split into two states $\sin(\Psi_z)$ and $\cos(\Psi_z)$. Other INS models related to land vehicles and robotics applications are presented in references [19,20]. An INS may compute the attitude using Euler angles, the direction cosine matrix or the quaternions. The quaternions method is advantageous since it requires less computation, gives better accuracy and avoids singularity [21]. Some INS error models using quaternions have been developed using the infinitesimal angle assumption [21–23]. Yu et al. proposed a quaternion error model for large attitude errors using rotation vector error, additive quaternion error and multiplicative quaternion error [24]. The model is linearised by solving additional three parameters. The author developed an INS error model for three large attitude errors and presented an algorithm for low cost IMU using psi-angle approach [7,8].

In this paper, an INS velocity error model, a position error model and an attitude error model are developed in the quaternion method in the computer frame. The velocity error propagation is presented using quaternions. Attitude errors are presented by the misalignment of the computer frame and the platform frame. Quaternion errors are solved in the computer frame. No small angle assumption is made in the model development. The models are suitable for both three small and large attitude errors cases. Based on the models developed in this paper, an INS algorithm is developed for low cost IMU.

Among external navigation sensors, global positioning system (GPS) can provide navigation information without drift over time. The integration of INS and GPS has been widely used for navigation [4–9]. In this paper, GPS is used for observation to aid the INS. The unknown INS attitudes are solved using in-motion alignment by GPS aiding. The Kalman filter is the most commonly used algorithm for fusing INS and other navigation data in both INS alignment and navigation phases. The Kalman filter is a linear statistical algorithm used to recursively estimate the states of interest [8,9,25,26]. It requires linear process model and linear observation model. The quaternions model developed in this paper is based on non-linear model. The extended Kalman filter (EKF) is the most widely used approach for a non-linear filter algorithm [8,9,25,26]. The EKF models are expanded using Jacobians. The implementation difficulty of Jacobians is one shortcoming of the EKF. This motivated the development of a new filter algorithm called the distribution approximation filter (DAF) [26,27]. In this paper, the DAF is implemented for the INS algorithm.

The paper will describe the notations and related definitions in Section 2. Section 3 addresses the development of generic INS error propagation models for large attitude errors in the quaternion approach using computer frame. Section 4 develops an INS algorithm for low cost IMUs in the computer frame using quaternions. Quaternions errors are exploited using the misalignment of the computer frame and the platform frame. The entire filter process model structure and the process noise are discussed. The DAF is used to implement this algorithm. The principle and the benefit of the DAF are briefly described. Section 5 presents the experimental results for the INS algorithm. The experimental results are presented to verify the quaternion models and the INS algorithms for low cost IMU. The experiment uses a low cost IMU aided by a differential GPS (DGPS). The results show how the heading errors are corrected from $\pm 180^\circ$ to $\pm 0.1^\circ$. Finally the conclusions are drawn.

2. Notations and definitions

body frame (b-frame)	frame fixed to the vehicle
computer frame (c-frame)	local level frame at the computed position
$C(Q_m^n)$	transformation matrix from m -frame to n -frame represented by quaternion Q_m^n
C_m^n	DCM from m -frame to n -frame
DAF	distribution approximation filter
DCM	direction cosine matrix
DGPS	differential GPS
earth frame (e-frame)	located at the earth centre
EKF	extended Kalman filter
GPS	global positioning system
I	3×3 identity matrix
IMU	inertial measurement unit
inertial frame (i-frame)	inertial space
INS	inertial navigation system
platform frame (p-frame)	frame which the transformed accelerations and angular rates from the accelerometers and gyros are resolved
phi-angle (Φ)	the angle between t-frame and p-frame
psi-angle (Ψ)	the angle between c-frame and p-frame ($\Psi = [\Psi_x, \Psi_y, \Psi_z]^T$)
Q_m^n	quaternion vector which represents the rotation from m -frame to n -frame
true frame (t-frame)	true local level frame at the true position
u^j	vector u resolved in j -frame
ε^j	vector of gyros drift error in j -frame
φ	the local latitude
ω_{kl}^j	angular rate between k -frame and l -frame resolved in j -frame
Ω_{kl}^j	skew symmetric matrix of ω_{kl}^j
∇^j	vector of accelerometers drift error in j -frame
$\otimes$	quaternion multiplication

Fundamental to the process of INS is the definition of coordinate frames [28]. Fig. 3 illustrates how t-frame fits into the earth frame. Fig. 4 shows the relationship among a number of coordinate frames that are defined in the above notations. The usage of these frames will be described in the following sections.

3. Development of INS error models using the quaternion approach

This section develops the INS error propagation models that are presented by quaternions. The models include the velocity error propagation model, the position error propagation model and the quaternion error propagation model. These models are developed in computer frame. The concepts of quaternions and coordinate systems (frames) are introduced.

3.1. Quaternions and coordinate systems

In the literature, to derive the INS error models, the phi-angle approach (or the true frame approach) and the psi-angle approach (or the computer approach) are used. The phi-angle approach perturbs the INS equations in the local level north-pointing Cartesian coordinate system that corresponds to the true geographic location of the INS.

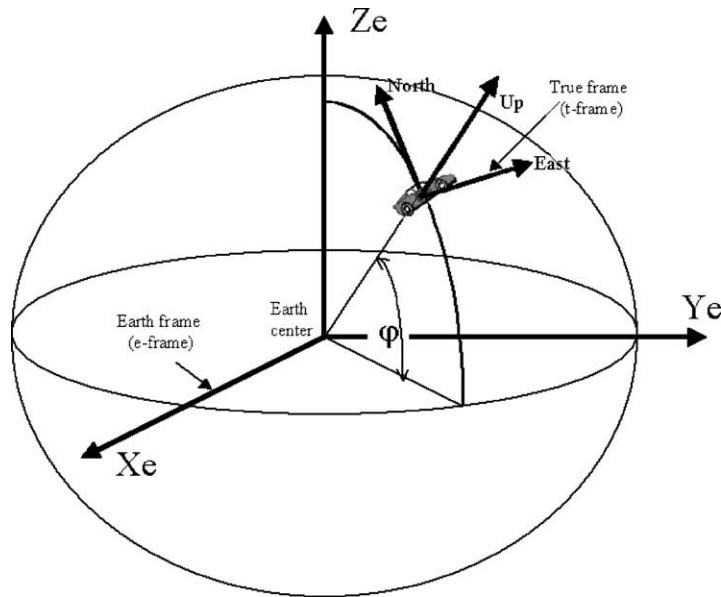


Fig. 3. t-Frame and the earth frame.

The psi-angle approach perturbs the INS equations in the computer frame which is the local level north-pointing coordinate system that corresponds to the geographic location computed by the INS, see Figs. 3 and 4. In this work, the computer approach is used. The INS error models for large attitude errors will be developed using quaternions. The coordinate systems used in these models are the computer frame (c-frame), the body frame

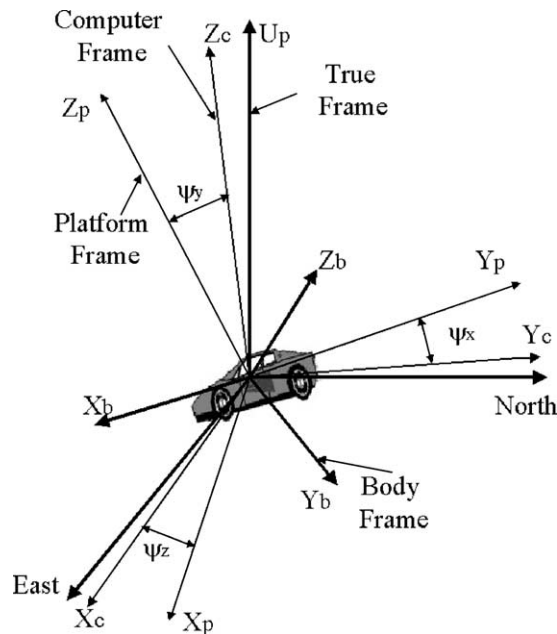


Fig. 4. Relationship of coordinate frames.

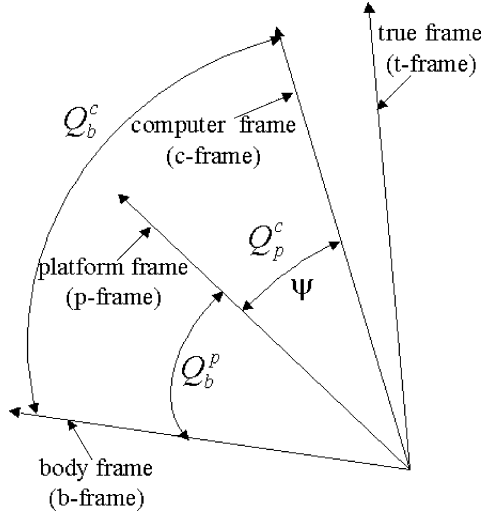


Fig. 5. Quaternions and frames.

(b-frame) and the platform frame (p-frame). The body frame is at the IMU whose axes are the IMU body axes, see Fig. 4.

A single rotation of a vehicle can be represented by quaternions. The quaternion attitude representation is a four-parameter representation based on the idea that a transformation from one coordinate frame to another may be affected by this rotation about a vector. The four elements of the quaternion are functions of this vector and the magnitude of the rotation [28,29]. Fig. 5 illustrates the relationship among the quaternions and the coordinate frames. The quantification of this relationship is presented as follows.

INS navigation computes the attitude transformation from the body frame to the platform frame. Quaternion $Q_b^p = [q_{b0}, q_{b1}, q_{b2}, q_{b3}]^T$ represents this rotation from the body frame to the platform frame [28,29]. Direct accelerometer outputs in the body frame are transformed to the platform frame by left multiplying the matrix $C(Q_b^p)$ to the accelerations in the body frame:

$$C(Q_b^p) = \begin{bmatrix} q_{b0}^2 + q_{b1}^2 - q_{b2}^2 - q_{b3}^2 & 2(q_{b1}q_{b2} - q_{b0}q_{b3}) & 2(q_{b1}q_{b3} + q_{b0}q_{b2}) \\ 2(q_{b1}q_{b2} + q_{b0}q_{b3}) & q_{b0}^2 - q_{b1}^2 + q_{b2}^2 - q_{b3}^2 & 2(q_{b2}q_{b3} - q_{b0}q_{b1}) \\ 2(q_{b1}q_{b3} - q_{b0}q_{b2}) & 2(q_{b2}q_{b3} + q_{b0}q_{b1}) & q_{b0}^2 - q_{b1}^2 - q_{b2}^2 + q_{b3}^2 \end{bmatrix}. \quad (1)$$

Quaternion Q_b^p is normalised in each computation at INS sampling time. The four elements satisfy the constraint [28,29]

$$q_{b0}^2 + q_{b1}^2 + q_{b2}^2 + q_{b3}^2 = 1. \quad (2)$$

The computer frame is the local level frame at the INS computed position, see Fig. 4. The rotation from the platform frame to the computer frame is represented by quaternion Q_p^c :

$$Q_p^c = [q_0, q_1, q_2, q_3]^T. \quad (3)$$

The transformation from the platform frame to the computer frame is given by the matrix $C(Q_p^c)$:

$$C(Q_p^c) = \begin{bmatrix} q_0^2 + q_1^2 - q_2^2 - q_3^2 & 2(q_1q_2 - q_0q_3) & 2(q_1q_3 + q_0q_2) \\ 2(q_1q_2 + q_0q_3) & q_0^2 - q_1^2 + q_2^2 - q_3^2 & 2(q_2q_3 - q_0q_1) \\ 2(q_1q_3 - q_0q_2) & 2(q_2q_3 + q_0q_1) & q_0^2 - q_1^2 - q_2^2 + q_3^2 \end{bmatrix}. \quad (4)$$

The transformation from the body frame to the computer frame is represented by the quaternion Q_b^c . This rotation can be considered as two consecutive rotations from the body frame to the platform frame, then to the computer frame. Thus,

$$Q_b^c = Q_p^c \otimes Q_b^p, \quad (5)$$

where $\otimes$ represents the quaternion multiplication. The transformation matrix $C(Q_b^c)$ is equal to the product of two transformation matrices:

$$C(Q_b^c) = C(Q_p^c) \otimes C(Q_b^p). \quad (6)$$

The inconsistencies of the platform frame and the computer frame cause the navigation quaternion error Q_b^p . Likewise the psi-angles which are the misalignment angles between the platform frame and the computer frame, quaternion Q_p^c consequently stands for the inconsistency of the p-frame and the c-frame. The attitude error model is therefore transferred to the modelling of the quaternion Q_p^c .

3.2. Velocity error propagation model

The following section develops the INS velocity error propagation model in the computer approach. The INS velocity is solved in the computer frame.

The true velocity vector $V_t^c = [V_x^c, V_y^c, V_z^c]^T$, in the computer frame is derived by the Coriolis equation [28,29]:

$$\dot{V}_t^c = f_t^c + g_t^c - (2\Omega_{ie}^c + \Omega_{ec}^c)V_t^c, \quad (7)$$

where f_t^c is the true specific force vector defined as the non-gravity force per unit mass exerted on the accelerometers [29]. f_t^c is resolved in the computer frame, g_t^c is the gravity vector resolved in the computer frame, Ω_{ie}^c the matrix of the earth rate resolved in the computer frame and Ω_{ec}^c the matrix representing the rotation from the computer frame to the earth frame. Ω_{ie}^c and Ω_{ec}^c are given by

$$\Omega_{ie}^c = \begin{bmatrix} 0 & -\omega_{ie} \sin(\varphi) & \omega_{ie} \cos(\varphi) \\ \omega_{ie} \sin(\varphi) & 0 & 0 \\ -\omega_{ie} \cos(\varphi) & 0 & 0 \end{bmatrix}, \quad \Omega_{ec}^c = \begin{bmatrix} 0 & -\frac{V_x^c \tan(\varphi)}{R} & \frac{V_x^c}{R} \\ \frac{V_x^c \tan(\varphi)}{R} & 0 & \frac{V_y^c}{R} \\ -\frac{V_x^c}{R} & -\frac{V_y^c}{R} & 0 \end{bmatrix}, \quad (8)$$

where φ is the local latitude. The earth rate ω_{ie} is equal to 7.272205×10^{-5} rad/s. R is the earth radius which can be considered as equal to 6,378,393 m. The INS velocity components V_x^c and V_y^c are computed in the computer frame in east and north.

Due to the existence of the error sources, the INS computes the following velocity vector:

$$\hat{V}_t^c = (f_t^p + \nabla^p) + \hat{g}_t^c - (2\Omega_{ie}^c + \Omega_{ec}^c)\hat{V}_t^c, \quad (9)$$

where $\hat{V}_t^c$ is the INS computed velocity vector in computer frame, f_t^p the true specific force vector resolved in the platform frame, ∇^p is the bias vector of the accelerometers in the platform frame and $\hat{g}_t^c$ the computed gravity vector.

Let $\Delta V^c = \hat{V}_t^c - V_t^c$ be the velocity error vector. The velocity error propagation model is derived by subtracting (7) from (9). Then,

$$\begin{aligned} \Delta \dot{V}^c &= \hat{V}_t^c - \dot{V}_t^c = ((f_t^p + \nabla^p) + \hat{g}_t^c - (2\Omega_{ie}^c + \Omega_{ec}^c)\hat{V}_t^c) - (f_t^c + g_t^c - (2\Omega_{ie}^c + \Omega_{ec}^c)V_t^c) \\ &= (I - C(Q_p^c))f_t^p - (2\Omega_{ie}^c + \Omega_{ec}^c)\Delta V^c + \nabla^p + \Delta g^c, \end{aligned} \quad (10)$$

where $\Delta g^c = \hat{g}_t^c - g_t^c$ and $f_t^c = C(Q_p^c) f_t^p$. The transformation quaternion matrix $C(Q_p^c)$ is shown in (4).

Therefore the velocity error model in quaternion form is

$$\Delta \dot{V}^c = (I - C(Q_p^c)) f_t^p - (2\Omega_{ie}^c + \Omega_{ec}^c) \Delta V^c + \nabla^p + \Delta g^c. \quad (11)$$

This model is applicable to the full range of misalignment angles without small angle errors assumption.

3.3. Position error propagation model

The INS computes position vector by integrating velocity vector. The velocity error directly impacts the position output. The position error is propagated over time. As discussed in [12], the position error model is de-coupled from the attitude errors. Therefore the position error model using quaternions has the same general form as in the psi-angle approach which is derived in [7,8]. The psi-angle and the quaternions do not appear directly in the position error equation. The position error model is as follows:

$$\Delta \dot{R}^c = -\Omega_{ec}^c \Delta R^c + \Delta V^c, \quad (12)$$

where ΔR^c is the position error vector resolved in the computer frame, ΔV^c the velocity error vector in the computer frame. Ω_{ec}^c is given by Eq. (8).

3.4. Quaternion error propagation model

In quaternion approach, the INS computed attitude vector is presented using quaternions. This section develops the quaternion error propagation model. During INS navigation, quaternion error is calculated to correct and update the navigation quaternions in real time at the INS sampling time.

The inconsistencies of the platform frame and the computer frame cause the navigation quaternion error. The quaternion error is modelled using Q_p^c which represents the misalignment of the platform frame and the computer frame, see Fig. 5.

The quaternion Q_p^c satisfies the following differential equation [28,29]:

$$\dot{Q}_p^c = \frac{1}{2} \Omega_{cp}^{*p} Q_p^c, \quad (13)$$

where the 4×4 matrix Ω_{cp}^{*p} is a function of ω_{cp}^p which is the rotation rate between the c-frame and the p-frame solved in the p-frame. ω_{cp}^p is presented by the change rate of the psi-angle $\Psi = [\dot{\psi}_x, \dot{\psi}_y, \dot{\psi}_z]^T$ which is the angle between the p-frame and the c-frame $\omega_{cp}^p = \dot{\Psi} = [\dot{\psi}_x, \dot{\psi}_y, \dot{\psi}_z]^T$.

Ω_{cp}^{*p} is presented by

$$\Omega_{cp}^{*p} = \begin{bmatrix} 0 & -\dot{\psi}_x & -\dot{\psi}_y & -\dot{\psi}_z \\ \dot{\psi}_x & 0 & \dot{\psi}_z & -\dot{\psi}_y \\ \dot{\psi}_y & -\dot{\psi}_z & 0 & \dot{\psi}_x \\ \dot{\psi}_z & \dot{\psi}_y & -\dot{\psi}_x & 0 \end{bmatrix}. \quad (14)$$

Extending (13):

$$\dot{Q}_p^c = \frac{1}{2} \begin{bmatrix} 0 & -\dot{\psi}_x & -\dot{\psi}_y & -\dot{\psi}_z \\ \dot{\psi}_x & 0 & \dot{\psi}_z & -\dot{\psi}_y \\ \dot{\psi}_y & -\dot{\psi}_z & 0 & \dot{\psi}_x \\ \dot{\psi}_z & \dot{\psi}_y & -\dot{\psi}_x & 0 \end{bmatrix} \begin{bmatrix} q_0 \\ q_1 \\ q_2 \\ q_3 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} -q_1 & -q_2 & -q_3 \\ q_0 & -q_3 & q_2 \\ q_3 & q_0 & -q_1 \\ -q_2 & q_1 & q_0 \end{bmatrix} \begin{bmatrix} \dot{\psi}_x \\ \dot{\psi}_y \\ \dot{\psi}_z \end{bmatrix}. \quad (15)$$

Let

$$B = \frac{1}{2} \begin{bmatrix} -q_1 & -q_2 & -q_3 \\ q_0 & -q_3 & q_2 \\ q_3 & q_0 & -q_1 \\ -q_2 & q_1 & q_0 \end{bmatrix}. \quad (16)$$

Then

$$\dot{Q}_p^c = B \begin{bmatrix} \dot{\psi}_x \\ \dot{\psi}_y \\ \dot{\psi}_z \end{bmatrix} = B\dot{\psi}. \quad (17)$$

The general psi-angle model when all the psi-angles are large is developed in [7,8]. It is given by

$$\dot{\psi} = (I - C_c^p)\omega_{ic}^c - \varepsilon^p, \quad (18)$$

where ε^p is the gyro error vector resolved in the platform frame, ω_{ic}^c is given by [29]

$$\omega_{ic}^c = \begin{bmatrix} 0 \\ \omega_{ie} \cos(\varphi) \\ \omega_{ie} \sin(\varphi) \end{bmatrix} + \begin{bmatrix} -\frac{V_y^c}{R} \\ \frac{V_x^c}{R} \\ \frac{V_x^c \tan(\varphi)}{R} \end{bmatrix}, \quad (19)$$

ω_{ie} being the earth rate, φ the local latitude and R the earth radius. The INS velocity vector $V_t^c = [V_x^c, V_y^c, V_z^c]^T$ is resolved in the c-frame.

The direction cosine matrix C_c^p which is the transformation matrix from the computer frame to the platform frame can be converted into the quaternions form as $C(Q_c^p)$:

$$C(Q_c^p) = \begin{bmatrix} q_0^2 + q_1^2 - q_2^2 - q_3^2 & 2(q_1q_2 + q_0q_3) & 2(q_1q_3 - q_0q_2) \\ 2(q_1q_2 - q_0q_3) & q_0^2 - q_1^2 + q_2^2 - q_3^2 & 2(q_2q_3 + q_0q_1) \\ 2(q_1q_3 + q_0q_2) & 2(q_2q_3 - q_0q_1) & q_0^2 - q_1^2 - q_2^2 + q_3^2 \end{bmatrix}. \quad (20)$$

Therefore, the quaternion error model could be derived from Eq. (17) and the psi-angle model (18):

$$\dot{Q}_p^c = B((I - C(Q_c^p))\omega_{ic}^c - \varepsilon^p). \quad (21)$$

That is

$$\dot{Q}_p^c = B(I - C(Q_c^p))\omega_{ic}^c - B\varepsilon^p. \quad (22)$$

Eqs. (16), (20) and (22) give the general quaternion error model without small angle error assumption.

4. INS algorithm for low cost IMU in quaternion approach

In the following sections, the INS error models developed in this work will be applied to develop an INS algorithm for low cost IMU.

It is argued that attitude transformation and update using quaternions can provide more accurate results than the direction cosine matrix and the Euler method [21–23]. Nowadays many INSs use quaternions. The INS algorithm developed in this work uses quaternions and the developed INS error models. INS navigation starts with its

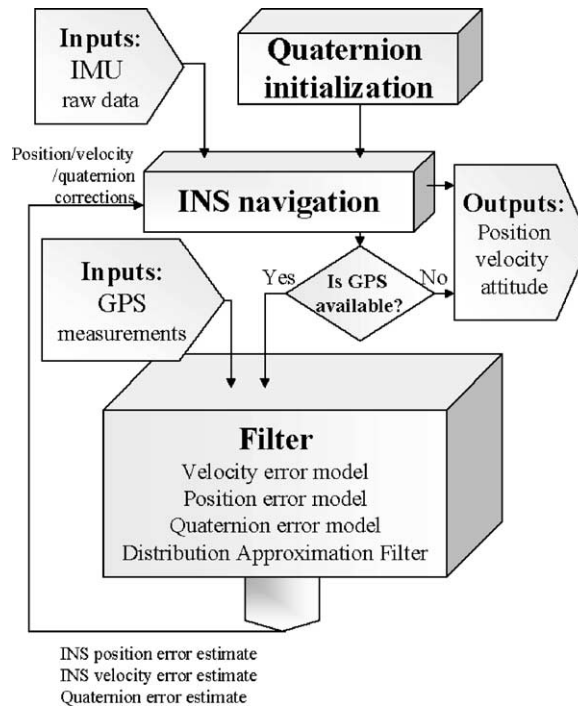


Fig. 6. Flow chart of the quaternion approach.

initial alignment phase, see Fig. 2. The inputs of initial alignment are initial velocity, position, attitude and initial quaternions. If an INS employs a low cost IMU, the initial attitude and initial quaternions are unknown due to the low precision of IMU outputs. Additional heading sensors could provide additional heading information to the navigation system, but this increases cost. Quaternion models for large attitude errors developed in this work can be used to solve the initial quaternion uncertainty without additional heading sensors, thus it reduces cost.

In this section, the quaternion approach in the computer frame is applied to the INS in-motion alignment to solve large azimuth uncertainty. The work flow of the algorithm is described. The algorithm is developed using a filter. The filter is composed of process model and observation model. The filter equations are given. These equations include velocity error equation, position error equation and quaternion error equation. The algorithm is discussed in detail in continuous time and discrete time. To implement the filter, the distribution approximation filter is used.

4.1. Algorithm work flow

The work flow chart of the algorithm is shown in Fig. 6. The inputs of the INS algorithm are the IMU raw data and GPS measurements. The INS navigation process starts from the quaternion initialisation. During the quaternion initialisation, the four elements of the quaternion vector are evaluated under the assumption of an arbitrary attitude. The INS computes the navigation data at each INS sampling time starting with the first quaternion vector. When a GPS fix is available, a filter based on the velocity, position and quaternion error models updates the states. The filter state vector consists of the INS velocity errors, the position errors and the quaternion errors in the computer frame. These estimates are used to correct the INS navigation and calibrate IMU biases. The outputs of the algorithm are INS computed position, velocity and attitude.

The computer frame is selected as the local level frame in east, north and up at the INS computed position. The filter details are presented as follows.

4.2. Filter equations

The filter consists of filter states, filter process model and filter observation model. This section presents the details of the filter equations.

4.2.1. Filter states

The filter propagates the estimated errors of the INS. The filter states in this algorithm are the INS velocity and position errors in the computer frame and the quaternion errors. The state vector of the filter is

$$X(t) = [\Delta V_x^c, \Delta V_y^c, \Delta V_z^c, \Delta R_x^c, \Delta R_y^c, \Delta R_z^c, q_0, q_1, q_2, q_3], \quad (23)$$

where $\Delta V^c = [\Delta V_x^c, \Delta V_y^c, \Delta V_z^c]^T$ is the INS velocity error vector in the computer frame, $\Delta R^c = [\Delta R_x^c, \Delta R_y^c, \Delta R_z^c]^T$ the INS position error vector in the computer frame and $Q_p^c = [q_0, q_1, q_2, q_3]^T$ the quaternion vector which represents the rotation between the platform frame and the computer frame.

4.2.2. Filter process model equations

The filter process model describes the filter state propagation over time. It contains velocity error equation, position error equation and quaternion error equation. The state vector process satisfies the following differential equation:

$$\dot{X}(t) = f_c(X, t) + G_c u(t), \quad (24)$$

where $f_c(X, t)$ is the process model, $u(t)$ the process state noise and G_c is called process noise input matrix [8,9,26,27]. The process model equations and the noise input matrix are derived from the INS error models developed as follows.

4.2.2.1. Velocity error equation. The velocity error model for large attitude errors in quaternion form is as Eq. (11). Most low cost INSs are used in level navigation applications and consequently the gravity error Δg^c in Eq. (11) can be ignored. The matrix $C(Q_p^c)$ transforms the true specific force vector $f_t^p = [f_x^p, f_y^p, f_z^p]^T$ in the platform frame to the computer frame. It is given by Eq. (4). Therefore the velocity error equation in the filter process model is

$$\begin{aligned} \Delta \dot{V}^c &= \begin{bmatrix} \Delta \dot{V}_x^c \\ \Delta \dot{V}_y^c \\ \Delta \dot{V}_z^c \end{bmatrix} = (I - C(Q_p^c)) f_t^p - (2\Omega_{ie}^c + \Omega_{ec}^c) \Delta V^c + \nabla^p \\ &= 2 \begin{bmatrix} q_2^2 + q_3^2 & -(q_1 q_2 - q_0 q_3) & -(q_1 q_3 + q_0 q_2) \\ -(q_1 q_2 + q_0 q_3) & q_1^2 + q_3^2 & -(q_2 q_3 - q_0 q_1) \\ -(q_1 q_3 - q_0 q_2) & -(q_2 q_3 + q_0 q_1) & q_1^2 + q_2^2 \end{bmatrix} \begin{bmatrix} f_x^p \\ f_y^p \\ f_z^p \end{bmatrix} \\ &\quad - \begin{bmatrix} 0 & -\left[\frac{V_x^c \operatorname{tg}(\varphi)}{R} + 2\omega_{ie} \sin(\varphi)\right] & \frac{V_x^c}{R} + 2\omega_{ie} \cos(\varphi) \\ \frac{V_x^c \operatorname{tg}(\varphi)}{R} + 2\omega_{ie} \sin(\varphi) & 0 & \frac{V_y^c}{R} \\ -\left[\frac{V_x^c}{R} + 2\omega_{ie} \cos(\varphi)\right] & -\frac{V_y^c}{R} & 0 \end{bmatrix} \begin{bmatrix} \Delta V_x^c \\ \Delta V_y^c \\ \Delta V_z^c \end{bmatrix} + \begin{bmatrix} \nabla_x^p \\ \nabla_y^p \\ \nabla_z^p \end{bmatrix}, \end{aligned} \quad (25)$$

where $\nabla^p = [\nabla_x^p, \nabla_y^p, \nabla_z^p]^T$ is the accelerometer error vector in the platform frame.

4.2.2.2. Position error equation. The position error vector in the computer frame is de-coupled from the quaternions. There are no differences in the position error model between the two cases whether the misalignments of the computer frame and the platform frame are small or large. Let the difference between the INS computed position and the true position in the computer frame ΔR^c be the position error in the computer frame. The position error equation becomes

$$\Delta \dot{R}^c = \begin{bmatrix} \Delta \dot{R}_x^c \\ \Delta \dot{R}_y^c \\ \Delta \dot{R}_z^c \end{bmatrix} = \Delta V^c - \Omega_{ec}^c \Delta R^c = \begin{bmatrix} \Delta V_x^c \\ \Delta V_y^c \\ \Delta V_z^c \end{bmatrix} - \begin{bmatrix} 0 & -\frac{V_x^c \operatorname{tg}(\varphi)}{R} & \frac{V_x^c}{R} \\ \frac{V_x^c \operatorname{tg}(\varphi)}{R} & 0 & \frac{V_y^c}{R} \\ -\frac{V_x^c}{R} & -\frac{V_y^c}{R} & 0 \end{bmatrix} \begin{bmatrix} \Delta R_x^c \\ \Delta R_y^c \\ \Delta R_z^c \end{bmatrix}. \quad (26)$$

4.2.2.3. Quaternion error equation. The attitude error here is modelled using the quaternion vector $Q_p^c = [q_0, q_1, q_2, q_3]^T$ which represents the misalignment of the platform frame and the computer frame.

The quaternion error equation can be written from Eq. (22):

$$\begin{aligned} \dot{Q}_p^c &= B(I - C(Q_p^c))\omega_{ic}^c - B\varepsilon^p \\ &= \begin{bmatrix} -q_1 & -q_2 & -q_3 \\ q_0 & -q_3 & q_2 \\ q_3 & q_0 & -q_1 \\ -q_2 & q_1 & q_0 \end{bmatrix} \begin{bmatrix} q_2^2 + q_3^2 & -(q_1q_2 + q_0q_3) & -(q_1q_3 - q_0q_2) \\ -(q_1q_2 - q_0q_3) & q_1^2 + q_3^2 & -(q_2q_3 + q_0q_1) \\ -(q_1q_3 + q_0q_2) & -(q_2q_3 - q_0q_1) & q_1^2 + q_2^2 \end{bmatrix} \begin{bmatrix} -\frac{V_y^c}{R} \\ \omega_{ie} \cos(\varphi) + \frac{V_x^c}{R} \\ \omega_{ie} \sin(\varphi) + \frac{V_x^c \operatorname{tg}(\varphi)}{R} \end{bmatrix} \\ &\quad - \frac{1}{2} \begin{bmatrix} -q_1 & -q_2 & -q_3 \\ q_0 & -q_3 & q_2 \\ q_3 & q_0 & -q_1 \\ -q_2 & q_1 & q_0 \end{bmatrix} \begin{bmatrix} \varepsilon_x^p \\ \varepsilon_y^p \\ \varepsilon_z^p \end{bmatrix}, \end{aligned} \quad (27)$$

where $\varepsilon^p = [\varepsilon_x^p, \varepsilon_y^p, \varepsilon_z^p]^T$ is the gyro error vector in the platform frame.

4.2.2.4. Filter process noise. Integrate the above filter equations into the filter process model (24), the process model becomes

$$\begin{bmatrix} \Delta \dot{V}^c \\ \Delta \dot{R}^c \\ \dot{Q}_p^c \end{bmatrix} = \begin{bmatrix} (I - C(Q_p^c))f_t^p - (2\Omega_{ie}^c + \Omega_{ec}^c) \Delta V^c \\ \Delta V^c - \Omega_{ec}^c \Delta R^c \\ B(I - C(Q_p^c))\omega_{ic}^c \end{bmatrix} + \begin{bmatrix} \nabla^p \\ 0_{3 \times 1} \\ -B\varepsilon^p \end{bmatrix}; \quad (28)$$

$f_c(X, t)$ in Eq. (24) is therefore given by

$$f_c(X, t) = \begin{bmatrix} (I - C(Q_p^c))f_t^p - (2\Omega_{ie}^c + \Omega_{ec}^c) \Delta V^c \\ \Delta V^c - \Omega_{ec}^c \Delta R^c \\ B(I - C(Q_p^c))\omega_{ic}^c \end{bmatrix}. \quad (29)$$

The filter process noise input matrix G_c in filter process model equation (24) is derived as follows. From Eq. (28), the process state noise $G_c u(t)$ is given by

$$G_c u(t) = \begin{bmatrix} \nabla^p \\ 0_{3 \times 1} \\ -B\varepsilon^p \end{bmatrix}, \quad (30)$$

where B is given by Eq. (16), $\nabla^p = [\nabla_x^p, \nabla_y^p, \nabla_z^p]^T$ and $\varepsilon^p = [\varepsilon_x^p, \varepsilon_y^p, \varepsilon_z^p]^T$ are the accelerometer and the gyro error vectors in the platform frame. They can be modelled as follows. Let $\nabla^b = [\nabla_x^b, \nabla_y^b, \nabla_z^b]^T$ and $\varepsilon^b = [\varepsilon_x^b, \varepsilon_y^b, \varepsilon_z^b]^T$ be the accelerometer and the gyro error vectors in the body frame. Then ∇^p , ∇^b , ε^p and ε^b satisfy the following transformation:

$$\nabla^p = C(Q_b^p) \times \nabla^b, \quad (31)$$

$$\varepsilon^p = C(Q_b^p) \times \varepsilon^b, \quad (32)$$

where the matrix $C(Q_b^p)$ transforms the vectors from the body frame to the platform frame. It must be noted that the quaternions Q_b^p in (31) and (32) do not contain the filter state estimate Q_p^c .

Define two transient matrices D^p and E :

Let

$$D^p = -B\varepsilon^p = \begin{bmatrix} D_1^p \\ D_2^p \\ D_3^p \\ D_4^p \end{bmatrix} \quad (33)$$

and

$$E = -B \times C(Q_b^p). \quad (34)$$

Then

$$D^p = -B\varepsilon^p = C(Q_b^p) \times \varepsilon^b = E \times \varepsilon^b. \quad (35)$$

The process state noise vector $u(t)$ in Eq. (24) becomes

$$u(t) = \begin{bmatrix} \nabla^p \\ D^p \end{bmatrix} = \begin{bmatrix} \nabla_x^p \\ \nabla_y^p \\ \nabla_z^p \\ D_1^p \\ D_2^p \\ D_3^p \\ D_4^p \end{bmatrix}, \quad (36)$$

$u(t)$ is a white Gaussian noise vector due to the following reason.

In INS alignment phase, the vehicle stays stable at its zero velocity stage. The turn-on biases of the accelerometers and the gyros in the body frame are removed. The remains of the bias ∇^b of the accelerometers and the bias ε^b of the gyros in the body frame are considered as white Gaussian noise. The linear combinations of jointly white Gaussian random variables are also white Gaussian random variables [30]. Therefore ∇^p , D^p and

$$\begin{bmatrix} \nabla^p \\ D^p \end{bmatrix}$$

are also white Gaussian noises.

$u(t)$ can be projected into the state space using the input matrix. G_c is therefore derived from Eqs. (30) and (36) as

$$G_c = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}. \quad (37)$$

4.2.3. Filter observation model

The filter observations satisfy the following observation model:

$$Z(t) = HX(t) + \omega(t), \quad (38)$$

where $Z(t)$ is the observation vector. The observation matrix H models the relationship between the observation vector and the state vector. $\omega(t)$ is the vector of observation noise.

In this algorithm, set observations as the position and the velocity information differences between the INS and the differential GPS. Then $Z(t)$ is given by

$$Z(t) = \begin{bmatrix} \hat{R}_{\text{INS}}^c - \hat{R}_{\text{GPS}}^t \\ \hat{V}_{\text{INS}}^c - \hat{V}_{\text{GPS}}^t \end{bmatrix} = \begin{bmatrix} \Delta R^c - \Gamma_{\text{GPS}} \\ \Delta V^c - \Theta_{\text{GPS}} \end{bmatrix} = \begin{bmatrix} \Delta R_x^c \\ \Delta R_y^c \\ \Delta R_z^c \\ \Delta V_x^c \\ \Delta V_y^c \\ \Delta V_z^c \end{bmatrix} - \begin{bmatrix} \Gamma_x \\ \Gamma_y \\ \Gamma_z \\ \Theta_x \\ \Theta_y \\ \Theta_z \end{bmatrix}, \quad (39)$$

where $\hat{R}_{\text{INS}}^c$ and $\hat{V}_{\text{INS}}^c$ are the INS computed position vector and velocity vector and $\hat{R}_{\text{GPS}}^t$ and $\hat{V}_{\text{GPS}}^t$ the GPS position vector and velocity vector outputs, respectively. $\Gamma_{\text{GPS}} = [\Gamma_x, \Gamma_y, \Gamma_z]^T$ is the white noise vector of the GPS position measurement and $\Theta_{\text{GPS}} = [\Theta_x, \Theta_y, \Theta_z]^T$ the white noise of the GPS velocity measurement. H is obtained by the relationship between the states and the measurement

$$Z(t) = H \begin{bmatrix} \Delta V^c \\ \Delta R^c \\ Q_p^c \end{bmatrix} - \begin{bmatrix} \Gamma_x \\ \Gamma_y \\ \Gamma_z \\ \Theta_x \\ \Theta_y \\ \Theta_z \end{bmatrix}. \quad (40)$$

H is therefore given by

$$H = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}. \quad (41)$$

4.3. The discrete-time filter

The above filter is developed in continuous time. The filter works at each discrete time called data fusion time. So the filter needs to be converted to a discrete-time filter. This section derives this transformation.

The filter accuracy is achieved by using a first-order Euler integration scheme at each data fusion time. In this algorithm the GPS sampling time is set as the data fusion time. Let dt be the interval of two consequent filter times t_{k+1} and t_k , $dt = t_{k+1} - t_k$, we have

$$\dot{X}(t_k) \approx \frac{X(t_{k+1}) - X(t_k)}{dt} \approx f_c(X(t_k), t_k) + G_c u(t_k). \quad (42)$$

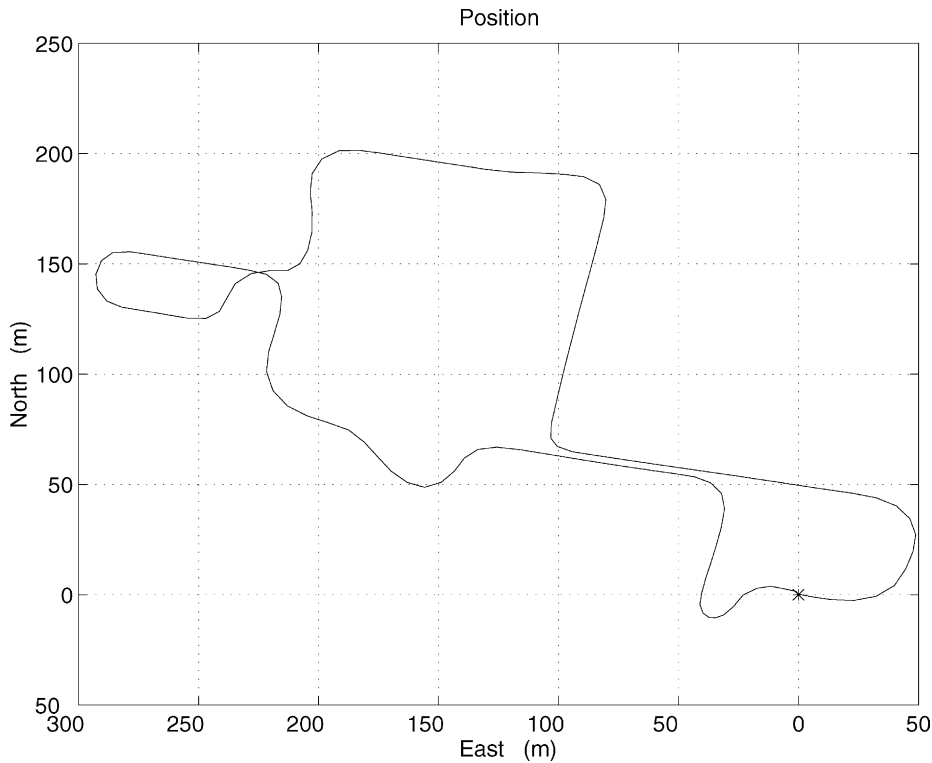


Fig. 7. The vehicle trajectory.

Therefore the discrete-time process model equation is

$$X(t_{k+1}) = X(t_k) + dt f_c(X(t_k), t_k) + G_c u(t_k) dt = f_k(X(t_k)) + G_k u_k, \quad (43)$$

where

$$f_k(X_k) = X(t_k) + dt f_c(X(t_k), t_k) = X(t_k) + dt \begin{bmatrix} (I - C(Q_p^c)) f_t^p - (2\Omega_{ie}^c + \Omega_{ec}^c) \Delta V^c \\ \Delta V^c - \Omega_{ec}^c \Delta R^c \\ B(I - C(Q_c^p)) \omega_{ic}^c \end{bmatrix} \quad (44)$$

and

$$G_k = G_c dt, \quad u_k = u_c(t_k). \quad (45)$$

The measurement is taken at the GPS sampling time. The discrete observation equation is

$$Z(t_k) = HX(t_k) + \omega(t_k). \quad (46)$$

For convenience, let $X_k = X(t_k)$, $Z_k = Z(t_k)$ and $\omega_k = \omega(t_k)$. The discrete filter is written as

$$X_k = f_k(X_k) + G_k u_k, \quad Z_k = HX_k + \omega_k. \quad (47)$$

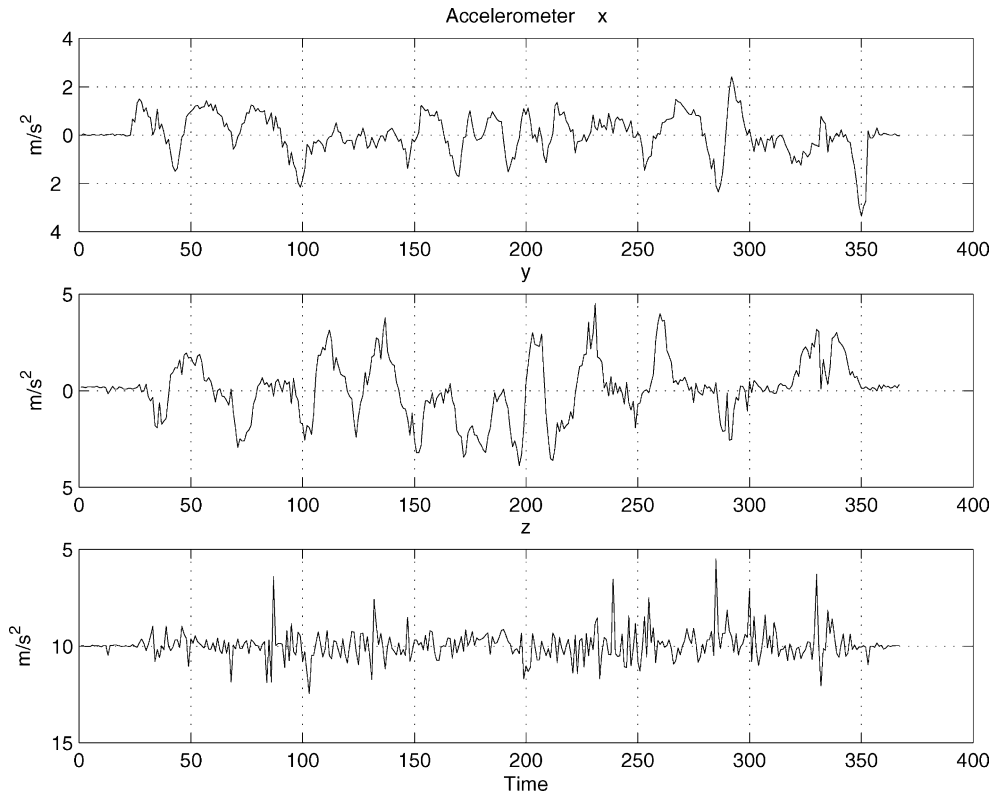


Fig. 8. Outputs of three accelerometers. The units for the x axis are seconds.

4.4. Filter implementation using DAF

In this section, step-by-step implementation of the algorithm developed above is presented. As in Eqs. (25)–(27), the filter process models of the quaternion approach are non-linear. The commonly used technology to solve non-linear filter as such is the extended Kalman filter [26,27]. EKF predicts the state of the system under the assumption that its process and observation models are locally linear. There are a number of problems with the EKF. The first is the need to analytically evaluate the Jacobian matrices [26,27] of the process and observation models. The Jacobian is not guaranteed to exist (e.g. at discontinuity), or might not have a finite value. Further, there can be considerable implementation difficulties when the system is composed of many states and is highly non-linear. Finally the linearisation can introduce significant errors [26].

These problems motivated the development of a new filter algorithm called the distribution approximation filter jointly by Julier et al. [26,27]. The DAF takes a mid-course between the analytical and numerical approaches. Like the numerical methods, it approximates the state distribution rather than the process or observation model.

The DAF uses the intuition that it is easier to approximate a distribution than it is to approximate an arbitrary non-linear function or transformation. A parameterisation could be found which captures the mean and covariance information while at the same time permitting the direct propagation of the information through an arbitrary set of non-linear equations. This can be accomplished by generating a discrete distribution having the same first and second order moments, where each point in the discrete approximation can be directly transformed. The mean and

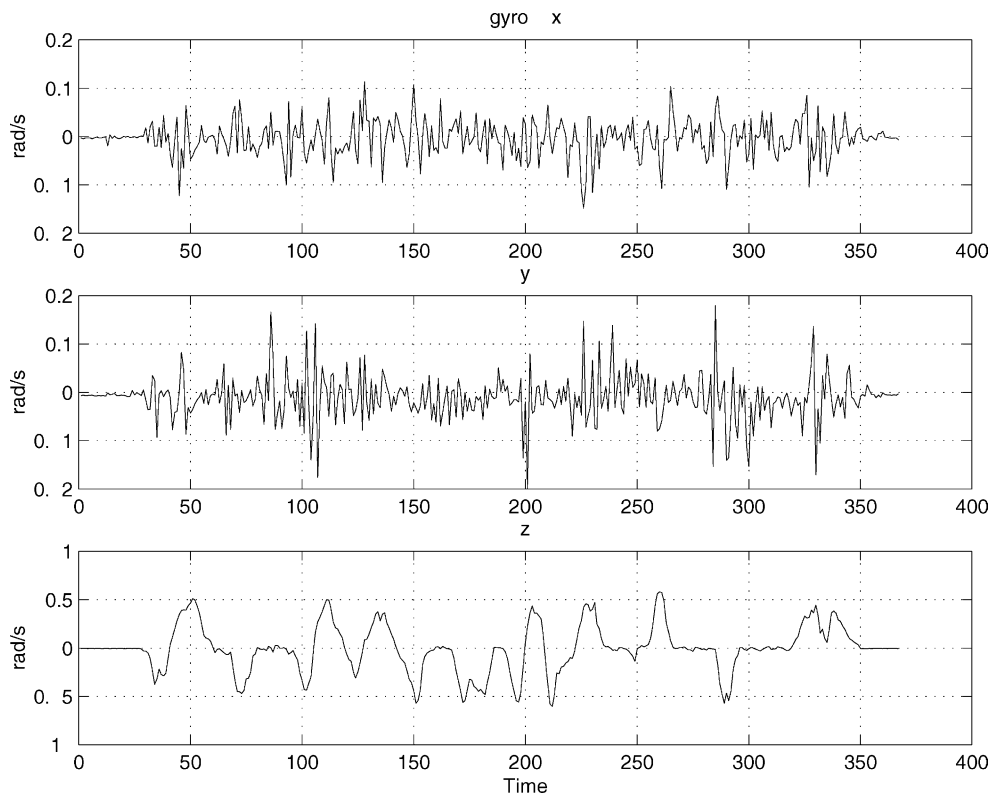


Fig. 9. Outputs of three gyros. The units for the x axis are seconds.

covariance of the transformed ensemble can then be computed as the estimate of the non-linear transformation of the original distribution.

The INS algorithm is implemented following the principle of DAF. The filter states are the errors of the INS velocity, position and quaternions. The discrete filter is as Eq. (47).

The initial filter state vector starts from X_0 with zero INS velocity error, zero INS position error and arbitrary initial quaternion error. The initial covariance matrix is set by P_0 . The diagonal terms of P_0 represent variances or mean-squared errors. The off-diagonal terms are set to be zeros.

The initial dynamic process noise is set as Q_0 , where the tuning of the diagonal terms of Q_0 determines the performance of the filter.

The initial measurement noise is R_0 , where the diagonal terms of R_0 are initial measurement noises on INS and GPS position and velocity.

The filter is implemented at time t_k using the DAF. The DAF procedure is as follows [26]:

- At the data fusion time t_{k-1} , the state estimate is X_{k-1} . The projected covariance matrix is P_{k-1} .
- At the current data fusion time t_k , compute the σ points using the projected covariance P_{k-1}

$$\sigma = \sqrt{nP_{k-1}}, \quad (48)$$

where n is the size of P_{k-1} .

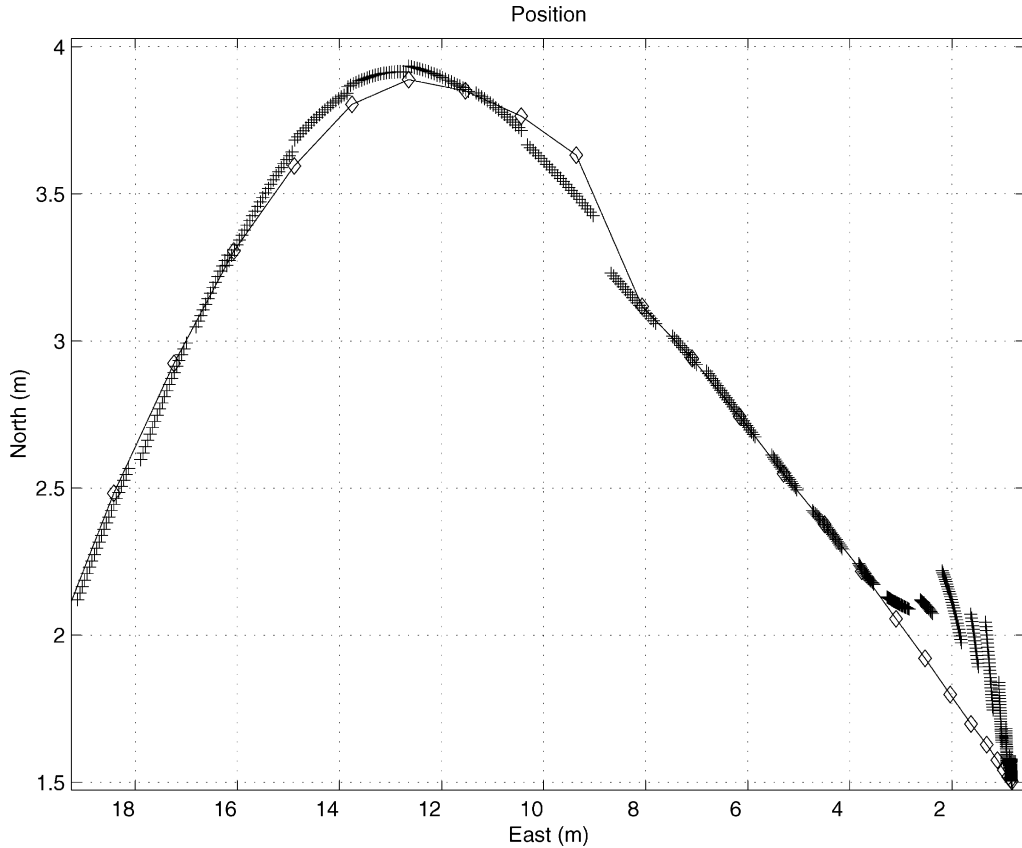


Fig. 10. At the beginning, the INS is uncertain within $\pm 180^\circ$. After 60 iterations of the filter, the INS heading error is corrected to less than $\pm 5^\circ$.

- Form the matrix χ_i

$$\chi_{ik,k} = [X_{k-1} + \sigma, X_{k-1} - \sigma]. \quad (49)$$

- Transform each point through the non-linear state process model as

$$\chi_{ik,k-1} = f_k(\chi_{ik,k}). \quad (50)$$

- Compute the predicted mean. Then the estimated errors are used to update the states. After the correction, the predicted mean state $X_{k,k-1}$ of the filter should be reset to zero.
- The predicted covariance $P_{k,k-1}$ is computed as

$$P_{k,k-1} = \frac{1}{2n} \int_{i=1}^{2n} \{\chi_{ik,k-1} - X_{k,k-1}\} [\chi_{ik,k-1} - X_{k,k-1}]^T + Q_k, \quad (51)$$

where Q_k is the dynamic noise covariance.

- Compute the filter gain K_k :

$$K_k = P_{k,k-1} H^T [H P_{k,k-1} H^T + R_k]^{-1} \quad (52)$$

with R_k being the measurement noise.

- Since the predicted state is zero, the update state estimate X_k is computed as

$$X_k = K_k Z_k, \quad (53)$$

where Z_k is the measurement of this time.

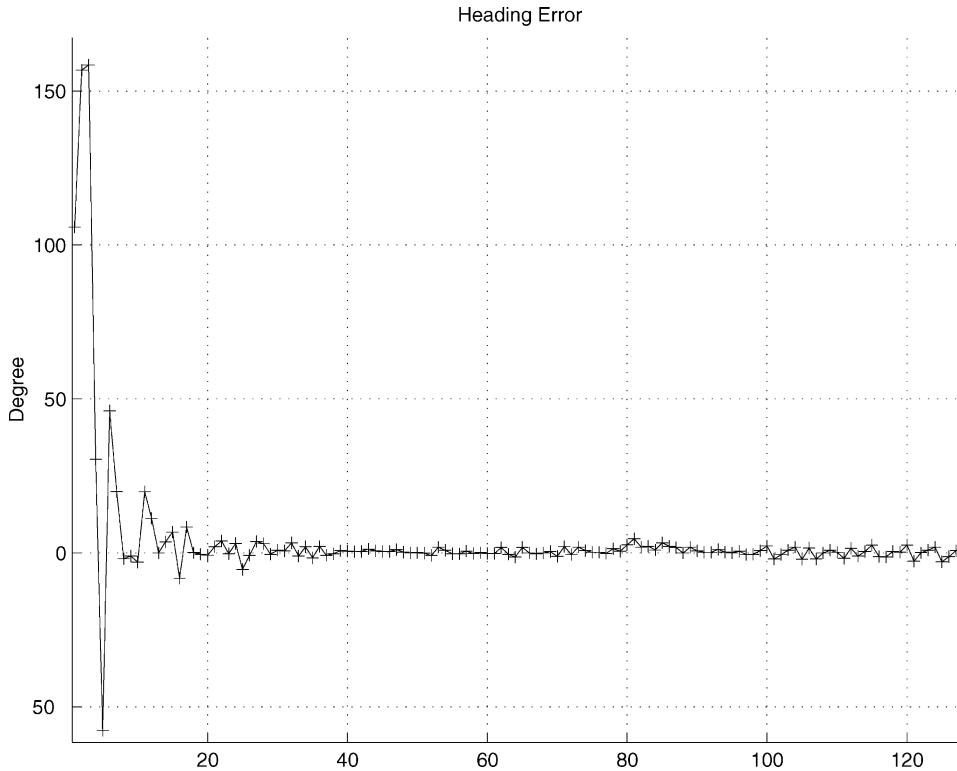


Fig. 11. The INS heading error is diminished to small angles after 60 iterations of the filter.

- The update covariance estimate P_k is computed by

$$P_k = [I - K_k H] P_{k,k-1}, \quad (54)$$

X_k is then used to correct the INS velocity, position and quaternions.

The quaternion correction is given by

$$Q_b^c = Q_p^c \otimes Q_b^p \quad (55)$$

with $\otimes$ being the quaternion multiplication.

After this filter cycle, the INS will continue to generate navigation data at the IMU sampling rate until the next GPS fix becomes available.

5. Experimental results of the quaternion algorithm

The quaternion algorithm developed in this paper is demonstrated using the experimental results in this section. The experimental platform consists of a low cost IMU aided by a DGPS. The experiment was conducted using a utility car in outdoor environment.

The IMU used in this experiment is a strapdown inertial measurement unit which contains three gyros, three accelerometers and two tilt gyros. The IMU is installed directly in the vehicle. The outputs of the IMU are in the

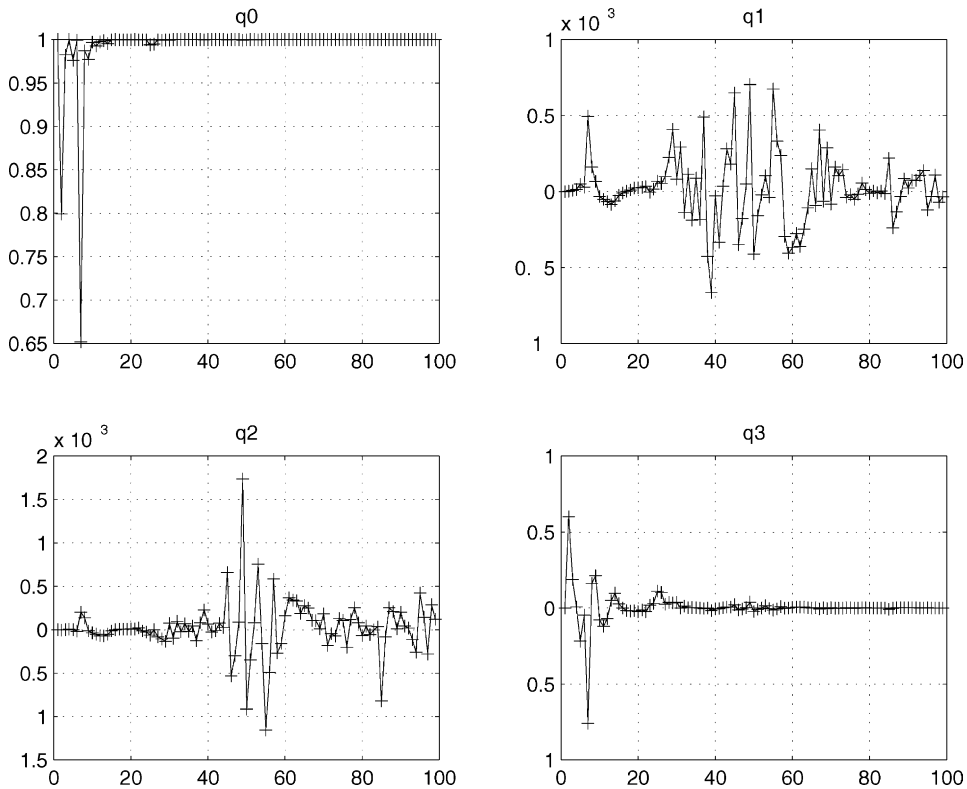


Fig. 12. Quaternions are corrected after each filter time.

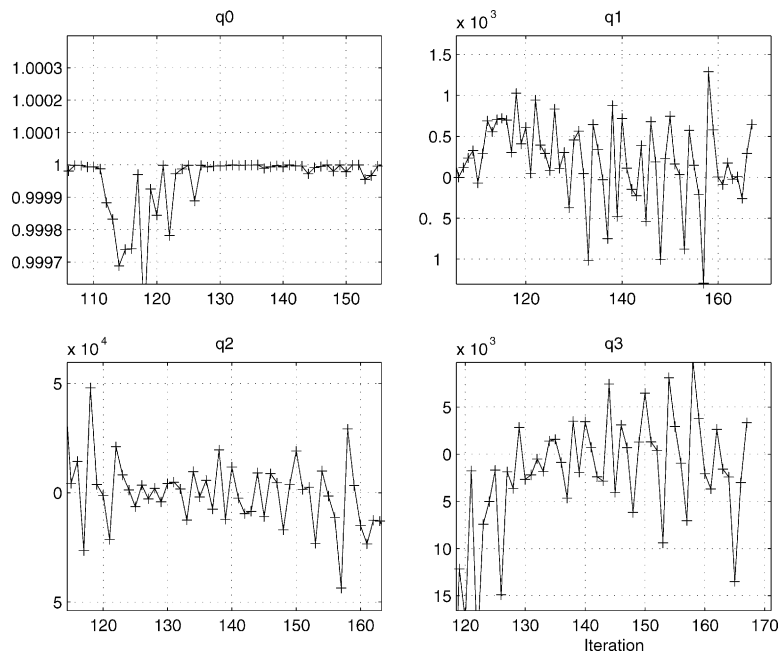


Fig. 13. The enhanced view of the quaternion errors between the filter iteration 100–160 which correspond to the vehicle running time 25–45 s. The four elements are very close to $[1, 0, 0, 0]^T$ which represents quaternions of a zero rotation from the computer frame to the platform frame.

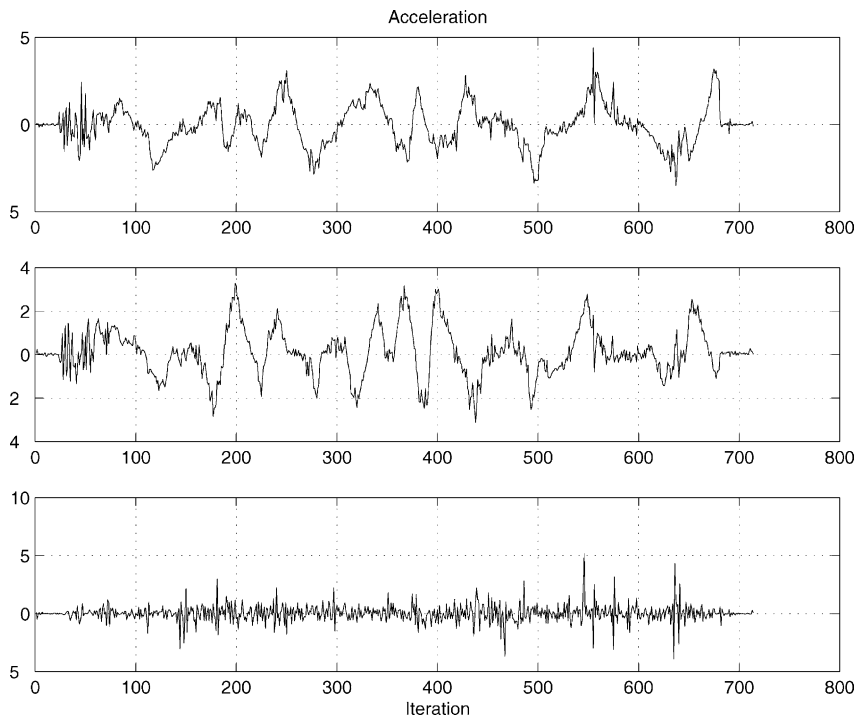


Fig. 14. Accelerations in the three axes of the platform frame. The unit of the plot is m/s^2 .

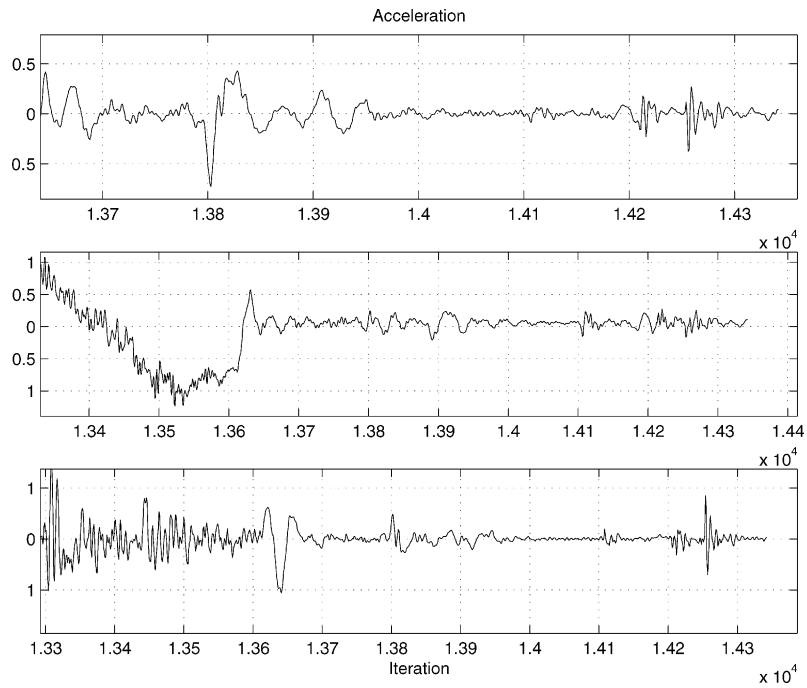


Fig. 15. At the end of the run, the vehicle is stationary. The accelerations in the three axes of the platform frame are very close to zero.

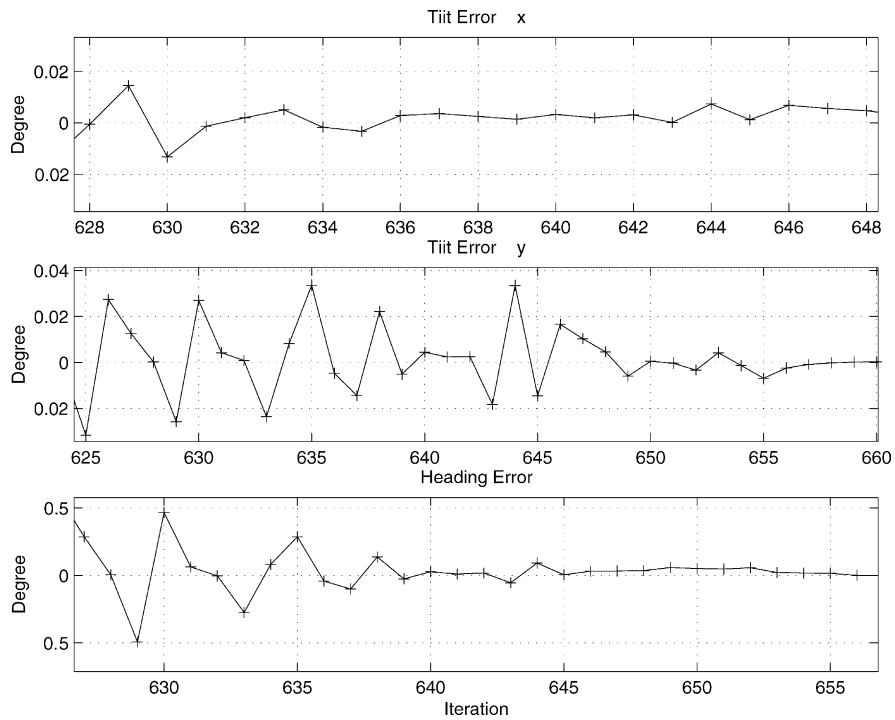


Fig. 16. Tilt errors are less than $\pm 0.04^\circ$. Heading errors are within $\pm 0.5^\circ$.

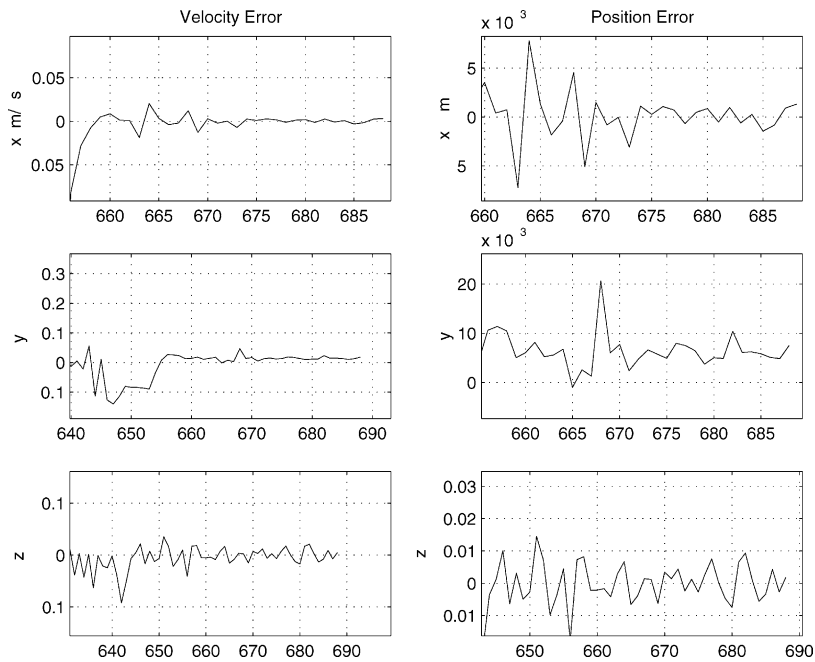


Fig. 17. The velocity errors are within 0.1 m/s. The position errors are within 0.02 m.

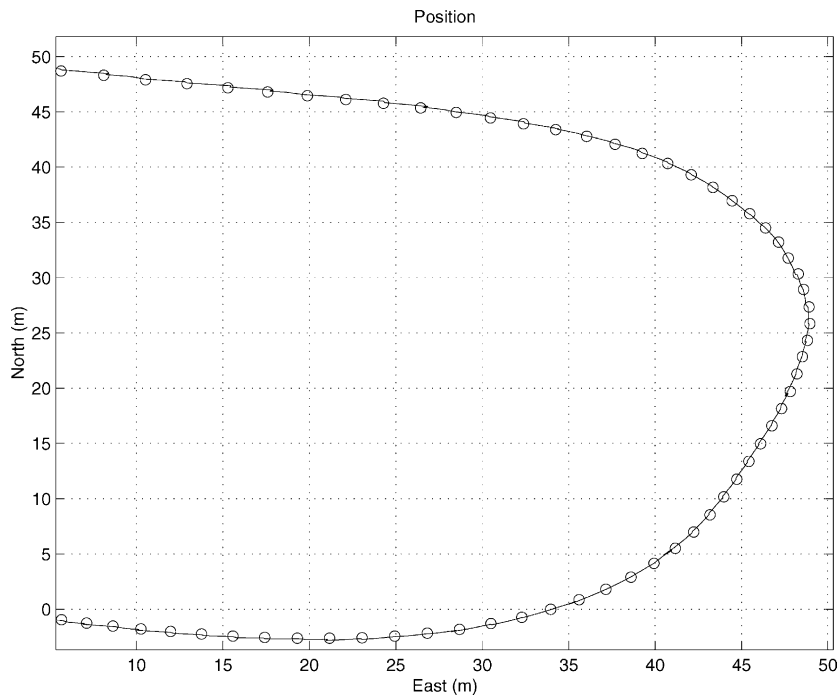


Fig. 18. The solid curve is the INS position output. Curve “o” is the GPS position output. INS outputs position data with the frequency of 84 Hz between two GPS position outputs.

body frame of the vehicle whose origin is defined at the IMU mass centre. The vehicle body frame is defined as the IMU body frame. Three accelerometers are called accelerometers x , y and z which are installed in the axes x , y and z of the body frame. The three gyros x , y and z which are installed in the axes x , y and z of the body frame provide the angular rate of the vehicle with respect to the body frame, see Fig. 1. The data sampling frequency is 84 Hz. The resolutions of the angular rate gyros and the accelerometers are 4.3115×10^{-4} rad/s and 0.0024 m/s², respectively. The resolution of the angular rate gyros is not enough to measure the earth rate.

The DGPS provides the position and velocity information in the WGS-84 reference with an accuracy of 2 cm CEP and 2 cm/s CEP, respectively. The sampling frequency of the DGPS is 10 Hz.

The vehicle trajectory in the experimentation is shown in Fig. 7. It starts from (0, 0) which is the origin of the local level frame. The vehicle moved from (0, 0) after a stationary period of approximately 12 s and returned and stopped at the starting position.

The outputs of the three accelerometers x , y and z are shown in Fig. 8. Fig. 9 is the plot of the raw data of the three gyros. They are all in the body frame.

The filter starts with arbitrary initial quaternions. The vehicle trajectory starts from (0, 0) of the local level frame which is at the right bottom corner of Fig. 10. The INS outputs wrong navigation information at the beginning of the run due to incorrect alignment. The position and heading correction can be appreciated in this figure after 60 filter iterations, which is about 20 s. The heading error is corrected to less than $\pm 2^\circ$. Fig. 11 shows this heading correction.

The quaternion errors are shown in Fig. 12. After 60 filter iterations which is approximately 20 s, the quaternion errors become very close to $[1, 0, 0, 0]^T$, which represents a zero rotation between the computer frame and the

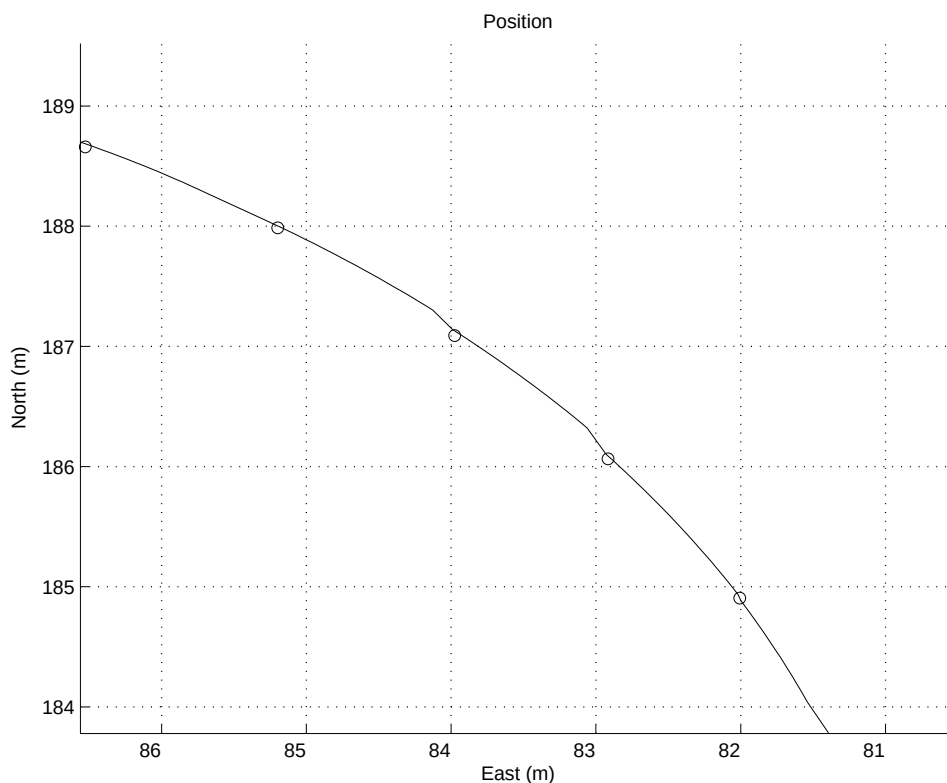


Fig. 19. INS position and heading corrections can be seen in the curve between two GPS position outputs.

platform frame. Fig. 13 is an enhanced view of the four elements of the quaternion errors at around 40 s of the vehicle running time.

The accelerations in the three axes of the platform frame are shown in Fig. 14. At the end of the run, the true acceleration in the platform should be zero. Fig. 15 is the enhanced view of the calculated acceleration in the platform frame. The acceleration in axes x and z are very close to zero. The error in axis y is about 0.08 m/s^2 .

The attitude errors in the three axes are shown in Fig. 16. The tilt errors are less than $\pm 0.04^\circ$. The heading errors are within $\pm 0.5^\circ$.

The velocity and position errors at the end of the run can be seen in Fig. 17. The velocity errors are within 0.5 m/s . The position errors are within 0.2 m .

Fig. 18 shows the position curves of the INS and GPS outputs after 170 s of the run. In this run the sampling frequency of the GPS is about 5 Hz. The INS position prediction matches the GPS position plot which is the label “o” in the plot.

Fig. 19 is an enhanced view of this position match. The INS position output is as the solid curve. The GPS position output is as the curve “o”.

6. Conclusion

Since the behaviour and the accuracy of an INS are strongly influenced by INS errors. Advanced INS algorithms are based on error modelling. Conventional INS solutions typically uses high accuracy IMUs which are rather expensive. The algorithm presented here removes its need for high accuracy IMUs and its need for additional heading sensor, therefore enables the use of low cost IMU to provide navigation solutions for field robotics.

This paper develops an INS error model for large attitude errors in the computer frame approach using quaternions. Differing from other quaternion models, quaternion errors are presented using the quaternions between the platform frame and the computer frame and this model makes no assumption of small angle errors.

Based on this model, an INS algorithm for low cost IMU using quaternions in the computer frame is designed. This algorithm deals with unknown initial attitudes using in-motion alignment aided by GPS measurement. To avoid deriving the Jacobian matrices of the non-linear filter, the distribution approximation filter is used in this algorithm.

The model and the algorithm are verified by experiments using real data. The experimental results using a low cost IMU and an aiding DGPS have shown that position, velocity and attitude accuracy can be achieved using the algorithm presented in this paper.

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References

- [1] E. Nebot, H. Durrant-Whyte, Initial calibration and alignment of low cost inertial navigation units for land vehicle applications, *Journal of Robotics Systems* 16 (2) (1999) 81–92.
- [2] S. Scheding, E. Nebot, M. Stevens, H. Durrant-Whyte, J. Roberts, P. Corke, Experiments in autonomous underground guidance, in: *Proceedings of the IEEE International Conference on Robotics and Automation*, Albuquerque, NM, USA, April 1997, pp. 1898–1903.
- [3] J. Guivant, E. Nebot, S. Baiker, High accuracy navigation using laser range sensors in outdoor applications, in: *Proceedings of the IEEE International Conference on Robotics and Automation*, vol. 4, April 24–28, 2000, pp. 3817–3822.
- [4] E. Nebot, H. Durrant-Whyte, High integrity navigation architecture for outdoor autonomous vehicles, *Journal of Robotics and Autonomous Systems* 26 (2–3) (1999) 81–97.

- [5] E. Nebot, Sensors used for autonomous navigation, in: *Advances in Intelligent Autonomous Systems*, Kluwer Academic Publishers, Dordrecht, 1999, Chapter 7, pp. 135–156.
- [6] S. Sukkarieh, Low cost, high integrity, aided inertial navigation systems for autonomous land vehicles, Ph.D. Thesis, The University of Sydney, 2000.
- [7] X. Kong, E. Nebot, H. Durrant-Whyte, Development of a non-linear psi-angle model for large misalignment errors and its application in INS alignment and calibration, in: *Proceedings of the IEEE International Conference on Robotics and Automation*, Detroit, MI, USA, May 1999, pp. 1430–1435.
- [8] X. Kong, Inertial navigation system algorithm for low cost IMU, Ph.D. Thesis, The University of Sydney, 2000.
- [9] S. Scheduling, High integrity navigation, Ph.D. Thesis, The University of Sydney, 1997.
- [10] D. Goshen-Meskin, I.Y. Bar-Itzhack, Unified approach to inertial navigation system error modeling, *Journal of Guidance, Control and Dynamics* 15 (3) (1992) 648–653.
- [11] I.Y. Bar-Itzhack, N. Berman, Control theoretic approach to inertial navigation systems, *Journal of Guidance and Control* 11 (3) (1988) 237–245.
- [12] D. Benson, A comparison of two approaches to pure-inertial and Doppler-inertial error analysis, *IEEE Aerospace and Electronic Systems* AES-11 (4) (1975) 447–455.
- [13] I.Y. Bar-Itzhack, Identity between INS position and velocity error models, *Journal of Guidance and Control* 4 (1981) 568–570.
- [14] I.Y. Bar-Itzhack, Identity between INS position and velocity error equations in the true frame, *Journal of Guidance and Control* 11 (6) (1988) 590–592.
- [15] T.M. Pham, Kalman filter mechanization for INS airstart, *IEEE AES System Magazine* 7 (January 1992) 3–11.
- [16] S.P. Dmitriyev, O.A. Stepanov, S.V. Shepel, Nonlinear filtering methods application in INS alignment, *IEEE Aerospace and Electronic Systems* AES-33 (1) (1997) 260–271.
- [17] B.M. Scherzinger, D. Reid, Modified strapdown inertial navigation error models, in: *Proceedings of the 1994 IEEE Position, Location and Navigation Symposium, PLANS'94*, Las Vegas, NV, USA, IEEE Aerospace and Electronic Systems, April 1994, pp. 426–430.
- [18] B.M. Scherzinger, Inertial navigation error models for large heading uncertainty, in: *Proceedings of the PLANS*, Atlanta, GA, USA, April 1996, pp. 477–484.
- [19] B. Barshan, H.F. Durrant-Whyte, Inertial navigation systems for mobile robots, *IEEE Transactions on Robotics and Automation* 11 (3) (1995) 328–342.
- [20] J. Vaganay, M.J. Aldon, Attitude estimation for a vehicle using inertial sensors, *Control Engineering Practice* 2 (2) (1994) 281–287.
- [21] H.K. Lee, J.G. Lee, Y.K. Roh, C.G. Park, Modeling quaternion errors in SDINS: computer frame approach, *IEEE Transactions on Aerospace and Electronic Systems* 34 (1) (1998) 289–297.
- [22] B. Friedland, Analysis strapdown navigation using quaternions, *IEEE Transactions on Aerospace and Electronic Systems* AES-14 (5) (1978) 764–768.
- [23] S. Vathsar, Optimal control of quaternion propagation in spacecraft navigation, *Journal of Guidance, Control and Dynamics* 9 (3) (1986) 382–384.
- [24] M. Yu, J.G. Lee, H.W. Park, Comparison of SDINS in-flight alignment using equivalent error models, *IEEE Transactions on Aerospace and Electronic Systems* 35 (3) (1999) 1046–1054.
- [25] C.K. Chui, G. Chen, *Kalman Filtering with Real-time Applications*, Springer-Verlag, New York, 1987.
- [26] S.J. Julier, Process models for the navigation of high speed land vehicles, Ph.D. Thesis, University of Oxford, 1997.
- [27] S.J. Julier, J.K. Uhlmann, H.F. Durrant-Whyte, A new approach for filtering nonlinear systems, in: *Proceedings of the 1995 American Control Conference*, Seattle, WA, 1995, pp. 1628–1632.
- [28] D.H. Titterton, J.L. Weston, *Strapdown Inertial Navigation Technology*, Peter Peregrinus Ltd., Herts., United Kingdom, 1997.
- [29] Z. Chen, *Principle of Strapdown Inertial Navigation System*, Beijing University of Aeronautics and Astronautics Press, Beijing, China, 1985.
- [30] P. Maybeck, *Stochastic Models, Estimation and Control*, Academic Press, New York, 1979.



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